Promising Less to Be Wrong Less Often: Four Renunciation Mechanisms as Alternative Actions in Crop Mapping

Anonymous authors

Affiliation withheld for double-blind review

Abstract. An automatic crop map that is unsure of a parcel can do more than answer anyway. It can narrow the catalogue it promises, decline to answer, hand back a small set of candidate labels, or fall back to a coarser agronomic category. These are four different ways of promising less, and the literature treats them as separate lines of work: the error–reject trade-off, set-valued prediction, catalogue granularity, and label hierarchies each have their own formulations, their own optima, and their own evaluations. Within the search reported in this paper, we found no comparison that reads them as alternative actions in one decision problem and asks which conclusions follow without choosing an outcome price. This paper sets up that comparison. We state the estimand and population before computing anything, hold every operating point to training and validation data only, and retain the complete six-coordinate loss space instead of inserting a price chosen by the authors. The data determine outcome-frequency vectors, uniform dominance, and linear switching boundaries; a stakeholder-specific preference does not follow unless an external loss point is supplied. The mechanisms, estimand, and class-distribution accounting are described here. The empirical verdict is deliberately withheld until the remaining inferential choices are frozen and the preregistered run is executed.

Keywords: Crop mapping · Selective classification · Set-valued prediction · Decision theory · Satellite image time series

1 Introduction

A parcel-level crop map is a promise. For every field in the scene it names a crop, and somebody downstream acts on that name: an inspector schedules a visit, an agronomist plans an input, an agency settles a subsidy. When the classifier is unsure, answering anyway transfers the uncertainty to that person without telling them.

There is more than one way to say less. The map can *shrink the catalogue* it is willing to promise and stop offering the classes it cannot separate; it can *abstain* on a parcel and hand it to a human; it can return a *set of candidate labels* and let the reader pick; or it can *fall back* to a coarser agronomic category that it can defend. Each of these buys accuracy with a different currency, and each one moves the cost somewhere else.

Promising less to be wrong less often is not a new idea. The error–reject trade-off is due to Chow [3], and Ha later generalized it to selecting a *subset* of classes—together with the optimality criterion that governs it [10]. Recent work gives efficient algorithms for the subset of maximum expected utility [14] and a systematization of the available formulations [4]. In crop mapping specifically, catalogue size has already been measured as a design variable [7], and the closest work to ours rejects pixels by uncertainty on the same benchmark and on three architectures also represented in our original candidate bank, making the trade-off between precision and recall explicit [16].

What we did not find is a comparison of these mechanisms *against each other*. Each line of work optimizes its own device and reports its own curve. Within the search scope described in Section 2, we found no work that puts catalogue shrinking, abstention, set-valued answers, and taxonomic back-off in one decision problem, preserves the complete loss space, and asks which conclusions are invariant to every admissible outcome price.

Contribution. We state the comparison and the conditions under which it can be believed. Concretely: (i) we cast the four mechanisms as alternative *actions* of the same decision problem and represent each by its complete outcome-frequency vector; (ii) we fix the estimand, the population, and the unit of analysis *before* computing anything, and we require the operating point of every mechanism to come from training and validation data alone; (iii) we derive uniform dominance and switching boundaries over the full normalized loss space, rather than choosing a favourable point; and (iv) we make the distribution of outcomes across classes a reported quantity rather than a side effect, because a mechanism that improves an average by silently dropping rare classes has not improved the same map.

What this paper does not claim. We do not infer a stakeholder's preferences from model outputs and do not choose a loss point on their behalf. The evidence can identify uniform dominance and the boundary at which one action becomes preferable to another; it cannot say where a particular inspector, adviser, or producer lies in that space. Any later external loss point is a declared sensitivity case, not a hidden input to the primary analysis. The recorded candidate search closed without an admissible confirmatory dataset, and its dataset-specific query log was not versioned; the resulting absence claim is therefore limited to the documented candidate set rather than the whole literature. The two available parcel datasets remain exploratory, and their preregistered run has not been executed.

2 Related Work

We organize this section by *limitation* rather than by application: for each line of work we state what it establishes and where it stops, because the stopping point is what defines the work that is left.

The renunciation framework already exists. The error–reject trade-off and its optimal rule are due to Chow [3]; that formulation is the degenerate case in which the only alternative to answering is silence. Ha generalized it to class-selective rejection, showing that selecting a subset of classes is the optimal compromise between error and the mean number of classes returned [10]; the treatment is decision-theoretic and assumes the posteriors are known. Mortier et al. give algorithms for the Bayes-optimal subset of maximum expected utility, with a utility that is a function of set size [14], and Chzhen et al. unify the existing set-valued formulations and their trade-offs [4], at infinite sample and under a plug-in principle. The framework is therefore not our contribution, and we say so on the first page. What these treatments share is that the price of an answer is a function of *how large* the answer is. They do not compare the four delivery actions through the same complete outcome-specific loss space.

Learned abstention is developed, and out of our scope by choice. Deep selective classification supplies risk guarantees [5], an architecture with an integrated reject head [6], a wagering-style loss [12], and an AUC-based selection criterion [15]. All of them abstain *per instance*; none emits a set or falls back to a coarser class. We do not reimplement them and we do not benchmark against them: our question is how abstention compares with the other three actions across a common loss space, not whether a better abstention rule exists.

Renunciation has reached Earth observation, and not through agriculture. The closest work to ours rejects pixels by an uncertainty threshold on PASTIS, using three architectures represented in our original candidate bank, and makes the trade-off explicit: not answering lowers recall and can raise precision when the rejected pixel would have been misclassified [16]. That is the risk–coverage compromise, performed and acknowledged. What it does not do is place the decision in the selective-classification frame, compare alternative actions, or characterize the loss-space boundaries at which the preferred action changes. Elsewhere in Earth observation, SHRUG-FM reports abstention with interpretable thresholds [9], on burn scars, floods, and landslides rather than on any agricultural task. Rejection methods have been applied to vegetation mapping [8], on airborne hyperspectral data rather than per-parcel satellite time series. Open-set crop recognition rejects the *unknown* through outlier exposure [2], which is a different question from distributing a degree of promise among known classes.

Catalogue size is already a measured design variable. Ghassemi et al. treat the number of classes as a design variable and measure it: starting from 52 LUCAS classes, a 26-class scheme balances accuracy against detail [7]. This is the direct precedent for reading catalogue shrinking as an operating point, and it forecloses any claim that we are the first to do so. There the setting is land cover with Random Forest; here it is one mechanism among four, represented in a common outcome vocabulary over the complete admissible loss space.

Taxonomic back-off already has a crop-mapping precedent. Multi-scale label hierarchies improve crop mapping from satellite image time series [17]. That work both predicts at three granularities and allows the output granularity to follow classification confidence; it is therefore a direct precedent for confidence-driven taxonomic back-off, not merely for hierarchical training. Hierarchical fusion also combines satellite, rotational, and contextual signals [1], but for information fusion rather than variable-granularity delivery. We do not claim the back-off mechanism as new. Hierarchical set-valued prediction has also been formalized with internal nodes as the usual valid sets [13]. The open comparison is whether this action is uniformly preferable to another action, and where their preference boundary lies otherwise.

Distributional effects are documented outside our domain. Selective classification can *magnify* disparities between groups: abstaining more can make outcomes worse for some of them [11]. The evidence is on demographic groups in vision and language, not on agronomic classes or parcels. We take it as the reason to report distribution across classes rather than as evidence about crops, and we note that the result runs opposite to the intuition that declining to answer is a neutral act.

Novelty, stated with its universe. We found no work that compares catalogue shrinking, per-parcel abstention, set-valued prediction, and taxonomic back-off *as alternative actions of the same decision problem*, reports their uniform-dominance order over the complete outcome-specific loss space, and accounts for how the underlying outcomes are distributed across classes. This statement is bounded by our search: arXiv, OpenAlex, and Semantic Scholar, 61 queries plus 6 manual ones, covering 2019–2026 together with the classics retrieved by name, in English only. That procedure indexes remote-sensing proceedings without DOIs poorly, and it is written in the vocabulary of selective classification—a limitation we observed in practice, since the closest work to ours [16] uses none of those terms and was found by another route.

3 Problem Statement and Estimand

3.1 Four Ways to Promise Less

Let $\mathcal{Y}$ be the class catalogue, let x_i be the observations for parcel i, and let $y_i \in \mathcal{Y}$ be its reference label. A conventional classifier commits to a map $x_i \mapsto \hat{y}_i \in \mathcal{Y}$: it must name exactly one class for every parcel, whatever its posterior looks like.

The mechanisms compared here all relax that commitment in the same formal way. Each replaces the point prediction with a subset of the catalogue,

$$C_m(x_i) \subseteq \mathcal{Y}, \qquad C_m(x_i) = \emptyset \text{ permitted}, \tag{1}$$

and they differ only in which subsets they may emit, and in what fixes the choice.

Catalogue shrinking. A subcatalogue $\mathcal{S} \subset \mathcal{Y}$ is fixed before any parcel is seen. A parcel whose unrestricted posterior maximizer lies inside $\mathcal{S}$ receives the restricted maximizer; a parcel whose maximizer lies outside receives nothing. The rule reads the posterior and never the reference label—keying delivery on the truth would make the mechanism unimplementable at inference time, and an earlier implementation in this line of work did exactly that. The decision is per class, and it is taken once.

Per-parcel abstention. The mechanism emits a singleton or the empty set, according to whether a confidence score clears an operating point. The decision is per parcel, and it is taken as many times as there are parcels.

Set-valued prediction. The mechanism emits a non-empty subset whose size varies with the difficulty of the parcel, calibrated so that a stated guarantee holds over the population. Every parcel receives an answer; what varies is how much the answer commits to.

Taxonomic back-off. The mechanism emits the set of leaves under a node of a class hierarchy. Formally this is a subset like any other; what distinguishes it is that the admissible subsets are those an agronomist can name, so the answer degrades to a coarser but still communicable category instead of to a list.

Two degenerate elements of the same space bound it: full-catalogue singleton prediction, which renounces nothing, and universal non-delivery, for which $C \equiv \emptyset$. Neither is one of the mechanisms, because neither chooses what to give up. They are not on the same footing, however, and Section 4 keeps them apart: the unmodified singleton is evaluated, as the reference against which every renunciation is priced, whereas universal non-delivery only fixes the price of silence and is never run. Writing all of these in one action space is this paper's first move, and it is what makes the question askable at all: as long as legend design, thresholding, conformal calibration, and hierarchical prediction are treated as separate techniques, there is no object on which to ask which of them one should prefer.

3.2 The Estimand

Fix a dataset d with its region, campaign, split, and predictor, and a mechanism m operating at τ_m. Each eligible held-out parcel falls into exactly one cell of a closed outcome account: correct precise delivery, incorrect precise delivery, non-delivery, a set that contains or excludes the truth, or a coarse answer that

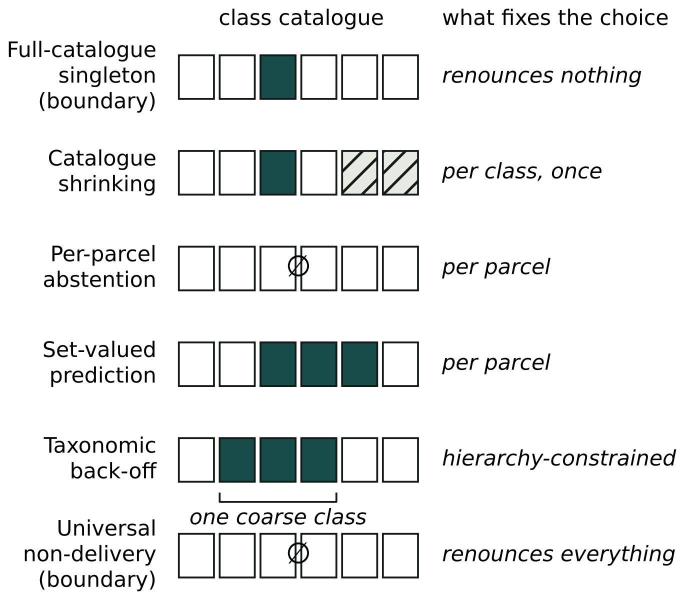

Fig. 1. The same action space. Every mechanism returns a subset of the class catalogue, with the empty set permitted; what separates them is which subsets they may emit and what fixes the choice. The two boundary rows are admissible actions rather than mechanisms under study. The figure is a schematic: it carries no measurement, and the number of cells stands for no particular benchmark.

contains or excludes the truth. Correct precise delivery is the zero-loss reference R_0; write the remaining cells as $L_1, \dots, L_6$. The empirical quantity is their frequency vector,

$$q_{d,m,j} \;=\; \frac{1}{N_d} \sum_{i=1}^{N_d} \mathbf{1}\{O_i(m) = L_j\}\,, \qquad j = 1, \dots, 6, \tag{2}$$

where N_d counts *all* eligible parcels of that block. For any finite non-negative loss vector, common positive scale is irrelevant to the ordering, so we normalize it to

$$\boldsymbol{\ell} \in \Delta^5 = \left\{ \boldsymbol{\ell} \in \mathbb{R}^6_+ : \sum_{j=1}^{6} \ell_j = 1 \right\}, \qquad R_{d,m}(\boldsymbol{\ell}) = \boldsymbol{q}_{d,m}^{\mathsf{T}} \boldsymbol{\ell}. \tag{3}$$

The data determine $\boldsymbol{q}$, not $\boldsymbol{\ell}$. We therefore report the full linear decision geometry instead of inserting a point. Arm m uniformly dominates arm n exactly when $q_{m,j} \le q_{n,j}$ for every cost-bearing cell and at least one inequality is strict. Otherwise, their switching boundary is the hyperplane $(\boldsymbol{q}_m - \boldsymbol{q}_n)^{\mathsf{T}} \boldsymbol{\ell} = 0$. A scalar switching price exists only inside a declared section that fixes the other coordinates. A two-dimensional picture is consequently a sensitivity view, not the complete five-dimensional space.

Cardinality is not the loss. Expected set size prices two sets of equal size identically even when one excludes the truth, and it assigns zero size to non-delivery. We retain it as a descriptor but never use it as the common objective. Nor do we infer a loss point from the model outputs: that would turn a modeling preference into a data result. A later externally sourced point may be overlaid as a non-confirmatory sensitivity case, with its provenance stated.

The population includes the parcels that receive nothing. N_d ranges over every eligible parcel in the block, including those the mechanism declines to answer and those whose class the shrunken catalogue no longer contains. Restricting the denominator to delivered parcels would price renunciation at zero and would score each mechanism on a population of its own choosing, so that a mechanism could improve its reported quality by answering less.

The operating point is fixed before the held-out block is touched. τ_m is derived entirely from training and validation. The held-out block estimates the realized outcome and nothing else. Re-equalizing loss, coverage, or delivery rate on the held-out block is prohibited, including as a post-hoc alignment intended to make mechanisms comparable: it is the leak that two audits of an earlier version of this work found, first in the threshold and then in its target rate.

3.3 Scope, Dependence, and What Is Not Claimed

Inference is *conditional on each dataset*, with its region, campaign, split, and predictor. The observational unit is the parcel. Parcels are not independent: they arrive in image patches that share acquisition, geography, and annotation, so the dependence cluster is the patch. The inferential bootstrap therefore resamples whole patches within spatial blocks and never parcels one by one. A separate paired interval over spatial-block means is reported beside it to expose between-block variation; because those folds share training data, it is a sensitivity view rather than evidence from independent territories. Below three unique paired units for either view, that view is reported descriptively — without interval, without p-value, and without multiplicity correction — because at that size the interval would describe the resampling scheme and not the data.

No quantity is pooled across datasets. Each dataset is reported on its own, and the second one serves as a replication attempt rather than as an additional sample of the first. The number of spatial blocks is a sensitivity parameter for the spatial structure of the estimate, not a count of replicates and not a lever on statistical power: partitioning the same territory more finely makes the resulting folds share more of their training data, so it produces the same evidence counted more often.

What this design gives up is stated here rather than in a limitations paragraph, because it constrains every claim that follows: **we do not claim transport**. Nothing reported for one region or campaign is asserted to hold in another. A conditional estimand is what the available partitions support; a marginal one would require a sample of regions that this work does not have, and the two were mixed in the design of an earlier version until an audit separated them.

The intended deployment context is Mexico, but our bounded search identified no public Mexican benchmark with parcel-level crop labels. The datasets available to this study therefore do not validate the deployment geography. This is a limitation of the evidence, not proof that such data do not exist, and no result below is generalized from the evaluated regions to Mexico.

4 Four Mechanisms as Alternative Actions

The comparison contains four mechanisms but six experimental arms. This distinction matters. There is one unmodified singleton baseline; catalogue shrinking has two arms, because the retained catalogue is ranked once by validation F-score and once by training support; and abstention, set-valued prediction, and taxonomic back-off contribute one arm each. Universal non-delivery is a boundary action used to identify the non-delivery cell, not a seventh experimental arm. Thus the count is

$$\text{baseline} + \text{two catalogue rules} + \text{three other mechanisms} = \text{six arms},$$

while the scientific question remains a comparison of four ways to renounce commitment. This also prevents a bookkeeping ambiguity in Section 3.1: the two boundary actions define the space, but they are not the two extra empirical arms.

4.1 A Common Output Type

Every arm returns the Boolean membership vector of $C_m(x_i)$ over the fixed catalogue. The representation contains no reference label, loss, or delivered-only filter. A row with one true entry is a precise delivery, a row with several true entries is an ambiguous or coarsened delivery, and an all-false row is non-delivery. Keeping the empty set inside the output type is important: it ensures that an abstained parcel remains in the population rather than disappearing before it can be counted.

The baseline emits the singleton containing the posterior maximizer. Both catalogue-shrinking arms apply the same operator to a different predeclared subcatalogue. If the unrestricted posterior maximizer belongs to that subcatalogue, it is emitted; otherwise the output is empty. The operator does not inspect the reference label. The two arms therefore differ only in the training-side rule that orders classes for withdrawal, not in how a test parcel is handled.

Per-parcel abstention emits the posterior maximizer when its confidence clears a fixed threshold and the empty set otherwise. Set-valued prediction sorts labels by posterior probability and emits the smallest non-empty prefix reaching a calibrated cumulative mass. The constructor applies that mass but never estimates it: calibration belongs entirely to training and validation, and any coverage guarantee depends on that preceding calibration rather than on the constructor alone.

The set-valued arm uses deterministic, non-randomized split-conformal adaptive prediction sets. For each calibration parcel, the score is the cumulative posterior mass through the true label under a stable probability ranking. The calibrated mass is selected by the exact finite-sample order-statistic rank $\lceil (n_{\mathrm{cal}} + 1)(1-\alpha)\rceil$, not by passing a rank fraction to an interpolating quantile API. If that rank lies beyond the calibration sample, the arm emits the complete catalogue rather than substituting the largest observed score. We choose the deterministic convention to avoid introducing an auxiliary randomization and seed; its possible conservatism is reported through observed marginal containment and set size. Held-out coverage is measured over all eligible parcels and may fall below the nominal level in a finite realization; such a result is reported and never used to recalibrate the mass.

Taxonomic back-off uses a predeclared fourth-edition HCAT cut over the leaf catalogue. A confident parcel keeps its singleton; an uncertain parcel receives every represented leaf below the node containing the posterior maximizer. The committed mapping resolves every leaf and cut node against the EuroCrops taxonomy before constructing a prediction, and rejects missing, repeated, non-ancestral, or out-of-catalogue assignments. Three cut nodes contain only one represented study leaf; back-off in those cases leaves the visible singleton unchanged rather than jumping post hoc to a broader node. Consequently, a malformed hierarchy cannot silently turn a parcel into non-delivery, and a deeper hierarchy would be a different operating rule rather than a choice made after observing outcomes.

All operating points—the retained catalogue, confidence threshold, calibrated mass, and coarse partition—are inputs to these operators. None is fitted by them, and none may be re-equalized on the held-out block. This separation is what makes test-label invariance inspectable: changing the test labels cannot change a delivered set.

Invariance to the held-out labels is a property of the operators; disjointness of the blocks is a property of whoever calls them, and the implementation no longer assumes it. A fitted operating point carries the stable parcel identities of the rows used to fit it and refuses to be applied to any of them. Identity is not inferred from posterior bytes: distinct parcels may have identical posteriors, and recomputing the same parcel may change its bytes. Handing the evaluated block back as calibration data therefore fails rather than passing quietly, which is how the corresponding leak entered an earlier version of this work.

The implementation makes this temporal order part of the interface. Its fitting function accepts training and validation arrays but has no held-out argument; its application function accepts held-out posteriors but has no label argument; only the evaluator receives labels, after all six membership matrices exist. A single orchestration path then composes these stages without supplying a default catalogue size, delivery target, miscoverage level, back-off rate, hierarchy, or loss point. This prevents an absent preregistration choice from becoming a library default.

The leaf catalogue is fixed by the training-side model output. Separately, a neutral evaluation universe is selected once from training labels using the predeclared support floor. Validation and held-out labels cannot add or remove a class from that universe. Every arm retains every held-out parcel in its outcome denominator, including non-deliveries and parcels whose true class lies outside the neutral classwise summary. A selected class absent from a held-out block remains visible with zero support and an undefined classwise statistic; it is not silently dropped or converted to a valid zero. Thus every primary frequency and classwise diagnostic uses denominators fixed before the arm's delivery is observed.

4.2 The Complete Action–Outcome Grid

The outcome interface follows the full grid fixed by the loss-space contract. A precise label is either correct, the zero-loss reference, or wrong. An empty set is non-response. A multi-label set either contains or excludes the truth. A coarse class likewise either contains or excludes the truth. These are seven cells in total. The two failure cells for non-singleton answers are indispensable: without them, a prediction set or a coarse answer would be charged only when it succeeds and would again acquire a free failure mode.

Classification into a cell follows the delivery visible to the user, not merely the name of the algorithm. A singleton emitted by the set-valued arm is scored as a precise label. A multi-label delivery is distinguished as an unconstrained label set or a named coarse class, because the two may have different consequences.

An empty row is non-response under every arm. Every parcel maps to exactly one cell.

The primary evaluator counts all seven cells and returns their complete frequency vector without accepting a loss point. A separate sensitivity interface accepts six non-negative finite losses, requires the correct precise label to remain the zero reference, and refuses an incomplete mapping. After removing common scale, these points form a five-dimensional simplex. Synthetic points test the necessary property that changing the relative price of a wrong label and silence can reverse which action has lower risk; they do not become empirical preferences.

Finally, the evaluator divides every cell count by the same number of eligible parcels, exactly as in Eq. (2). Non-deliveries therefore contribute their count to the same denominator as every other outcome. Risk at any declared sensitivity point is the dot product in Eq. (3). Expected set size is reported separately as a descriptor. The earlier size-based utility remains useful for a diagnostic link to macro recall, but it is not the common currency used to rank the six arms.

5 Evaluation Protocol

The protocol is executable without a point loss. It first estimates one complete outcome-frequency vector per arm, then derives dominance and switching geometry over every admissible normalized loss. No empirical contrast was run in order to write this section.

5.1 Split Order and Pair Formation

For each dataset and eligible predictor, training fixes the neutral class universe and the support-based catalogue; validation fixes the remaining operating-point components; and the held-out block is touched only after those choices are immutable. Every parcel carries a stable identity across these stages. Training and validation identities must be disjoint, and applying a fitted operating point to any identity seen in either split is an error. This check uses parcel identity rather than posterior equality, which avoids both accepting a recomputed copy of a parcel and rejecting two distinct parcels whose posteriors happen to coincide.

The common catalogue size is the smallest prefix of classes ordered by training support whose cumulative training mass reaches 0.9, with class id breaking support ties; the F-score and support rules use that same size. Validation targets a 0.9 delivery rate for abstention, a 0.1 miscoverage level for the adaptive set, and a 0.9 precise-leaf rate for taxonomic backoff. A class enters the neutral universe with at least 20 training parcels. These are author-chosen nominal intervention points, not equal-loss points, and none may be rematched on the held-out split.

The evaluator then assigns every eligible held-out parcel under every arm to exactly one of the seven closed outcome cells. Pairing is by parcel identity, not row position: one arm may be reordered, but it may not omit, duplicate, or substitute a parcel. Its spatial-block and `patch_id` metadata must agree with the comparator for that identity. The training-derived class universe travels with the outcome profile and cannot be reconstructed from a held-out sample or a bootstrap draw.

Evidential regime is carried by the artifacts rather than inferred from their filenames. A closed protocol inventory covers every file under the new result root. Each empirical CSV row or flat JSON record repeats exactly one declared regime, and the custody row repeats the same value. A non-empirical exception is limited to sealed PDF or SVG design diagnostics directly under the figure directory; neither a different location nor a symbolic link qualifies. A confirmatory result must be held out, identify the frozen confirmatory dataset, and descend in Git from the preregistration commit. The current inventory is explicitly mechanics-only and contains no empirical result; it therefore records absence rather than turning an empty table into evidence.

The third-dataset search is closed under a narrower, explicit boundary. Its register contains 14 documented candidate entries across two screening rounds, all adjudicated before any supplemental search or empirical run. None meets every structural admission criterion. Because no dataset-specific query log was versioned, this is not an exhaustive-search claim: it supports only the statement that no admissible confirmatory dataset was found in the recorded candidate set by the declared cutoff. The eligibility criteria are author-chosen scope and reproducibility rules, not intrinsic mathematical constraints on crop mapping. In particular, the label-cardinality floor, geography, dataset-supplied spatial split, and direct-download requirement could be chosen differently in another study. The empty branch is fixed in advance: there is no confirmatory stratum; PASTIS and BreizhCrops remain exploratory; and reopening requires a versioned amendment committed before a new candidate is reviewed.

5.2 Two Declared Uncertainty Views

For each arm pair and cost-bearing cell, the contrast is the left-arm indicator minus the comparator indicator. The primary, patch-cluster view is centred on the six paired frequency differences in Eq. 2. It samples whole `patch_id` groups with replacement within each block, carries every parcel and every arm of a sampled patch together, and combines the resampled block totals using the original parcel shares of those blocks. There is no parcel-level bootstrap.

Separately, the six indicator differences are averaged within each spatial block and those block means are given equal weight. A paired Student interval over one declared coordinate or declared linear section exposes between-block variation; it does not turn overlapping folds into independent regions. This is explicitly a sensitivity estimand: when block sizes differ, its centre need not equal the parcel-population contrast. Forcing both centres to coincide would silently replace Eq. 2 with an equal-block target.

Both views retain the complete parcel denominator and are reported side by side. If a view has fewer than three unique paired units, the implementation returns the observed block deltas and a reason, but no interval, no p-value, and no eligibility for a Holm family. A comparison declared identical by construction before evaluation is retained as a degenerate self-check and explicitly excluded from multiplicity rather than inflating the family.

5.3 Spatial Partition Selection and Sensitivity

The number of spatial blocks is selected before arm outcomes are evaluated. The rule sees only the exact minimum test-to-training centroid distance and the buffer used to construct the folds: the former must be at least one order of magnitude larger than the latter. This is a declared design threshold, not a proof of spatial independence; centroid distance is an upper bound on boundary separation, and residual spatial autocorrelation is not diagnosed here. Under the sealed profile, the rule uniquely selects five blocks. If no candidate or more than one candidate passes, the implementation stops instead of inventing a smallest- or largest-k tie-breaker.

An earlier rule also required a minimum number of supported classes in the worst block. That condition belonged to a retired classwise macro F-score estimand. It cannot identify the current mean loss over all eligible parcels and therefore remains a composition diagnostic rather than a selection condition. This distinction prevents a criterion made obsolete by the estimand from surviving merely because it still selects the same value.

Selection and sensitivity are separate interfaces. After the partition is fixed, every declared k is applied to the same paired outcome indicators and the entire curve is reported. Partition fingerprints are invariant to row order and arbitrary block names, while duplicate k values, omitted or added values, construction buffers outside the declared design, effective block counts that differ from the requested value, and adjacent parcel universes are errors. Finer partitions thus expose sensitivity to spatial grouping; they never enter the analysis as additional independent evidence or as a route to statistical power.

5.4 Power Has the Same Unit as Its Test

The minimum detectable effect for the paired block analysis is obtained from the noncentral Student distribution, not by adding two central-t quantiles. For n independent paired blocks, degrees of freedom $\nu = n - 1$, assumed standard deviation σ of the paired block differences, and test-wise level α^*, let

$$c = t_{1-\alpha^*/2,\nu}, \qquad \lambda = \frac{\delta\sqrt{n}}{\sigma}. \tag{4}$$

The implementation solves for the smallest absolute δ such that

$$\Pr\{T_\nu(\lambda) < -c\} + \Pr\{T_\nu(\lambda) > c\} \geq 1 - \beta. \tag{5}$$

All inputs are explicit. The planning standard deviation carries an outcome-independent source; it is not estimated from the contrast whose detectability is being assessed. Likewise, a hypothetical unit count is evaluated against a predeclared planning effect, never the observed delta. Repeated unit identifiers cannot increase n, fewer than three units are not assigned a publishable MDE under our declared rule, and zero observed variance is not converted into a zero planning scale. For an unadjusted comparison $\alpha^* = \alpha$. For a planned Holm family with m comparisons, prospective reporting uses the first-step threshold $\alpha^* = \alpha/m$; it is not described as one universal Holm level for every ordered test.

This calculation belongs only to the equal-block sensitivity estimand above. It does not describe the primary patch-cluster analysis, whose power depends on prospective outcome profiles and the cluster structure and is evaluated by simulation. Calling the block MDE the power of the primary estimand would recreate the unit mismatch that the two-view protocol was introduced to prevent. The code is executable on synthetic inputs, but no empirical MDE is reported here.

5.5 Dominance, Boundaries, and Multiplicity

Every stratum contains all unordered arm pairs and all six cost-bearing outcome coordinates. The primary scalar for an ordered pair is

$$D_{m,n} = \max_{j=1,\ldots,6} \left(q_{m,j} - q_{n,j}\right). \tag{6}$$

Observed componentwise dominance is descriptive. An inferential dominance claim requires a simultaneous one-sided upper bound for $D_{m,n}$ below zero, obtained from a maximum statistic over the complete predeclared family. Pairs or coordinates are not removed after their estimates are seen. If a required comparison is missing or has too few paired clusters, the affected simultaneous claim is withheld and the observed vectors remain available as descriptive output.

The frozen primary procedure uses a family-wise alpha of 0.05, 95% intervals, 9,999 paired patch-cluster resamples within spatial block, add-one bootstrap probabilities, and seed 20260909. Within each confirmatory stratum, the critical value is the maximum absolute studentized centred deviation over all 15 unordered arm pairs and 6 cost coordinates: 90 coordinates, or 180 directional inequalities. A zero standard error withholds the inferential claim. Studentization is performed inside every resample with that resample's own block-centred patch-cluster standard error; reusing one denominator for all resamples is prohibited. A relation stated across predictors uses an intersection–union rule and must hold in the same direction in every predeclared predictor stratum; no favourable predictor is selected.

The fitted winner region of arm m is the intersection of the simplex with the half-spaces $(\boldsymbol{q}_m - \boldsymbol{q}_n)^\mathsf{T}\boldsymbol{\ell} \leq 0$ for every comparator n. The boundary for one pair is the corresponding equality. These objects use all coordinates and do not require a probability measure over the simplex. A plotted two-dimensional section must declare the coordinates it fixes; it cannot be labelled as the full map or converted into a universal scalar threshold.

The separate class-distribution analysis uses the class-unweighted mean absolute deviation of promise retention. For a catalogue arm, class retention is one exactly when the true leaf belongs to its training-fitted legend; for abstention it is the class-conditional rate of nonempty delivery. The two planned contrasts are F-score catalogue minus abstention and support catalogue minus abstention, with positive direction and one Holm family of size two. If any class in the training-fixed common universe is absent, the whole H2 family is withheld rather than changing the denominator. Observed equality does not establish equivalence.

Point losses remain available only for explicit, non-confirmatory sensitivity cases. Such a point must name an external source or a declared geometric section; it never selects the operating point, defines the primary family, or licenses a claim about a stakeholder group. Ratios and Fieller sets implemented for the earlier point-loss design are outside the primary analysis unless a later amendment identifies their numerator, denominator, unit, and source before evaluation.

The target power is 0.8. Its sealed prospective alternative moves one percentage point out of each of the six adverse cells into the zero-loss reference and rotates the advantaged position through all six arms. Candidate admission uses 999 simulated datasets on the planned patch and block skeleton, the same seed and primary bootstrap, and intracluster correlations 0, 0.05, 0.1, and 0.2. The minimum one-sided 95% Wilson lower bound over arm placements and the correlation grid must reach the target. This author-chosen alternative is neither an observed effect nor a practical-importance margin. These values are frozen before empirical execution; this section still contains no empirical result.

6 Results

7 Discussion

Data and Code Availability

The anonymous submission archive contains the complete LaTeX project used to build this manuscript, including its bibliography, generated protocol appendix, and figures. It contains no empirical result artefacts because the comparison has not yet been run. The analysis code and derived artefacts for any eventual empirical results are maintained privately during double-blind review and are not part of this source archive; they will require a separate, citable release before an empirical claim can be published. Source satellite data are not redistributed and remain subject to the access and licence terms of their original providers.

A Frozen Analysis Contracts

This appendix exposes the decisions that the empirical run is not allowed to choose. It contains protocol structure and artifact identities only—no observed loss, model ranking, or empirical contrast. The source is generated from the versioned contracts; a missing, unsealed, or byte-divergent contract stops the build.

Estimand and inferential units. Inference is conditional on each dataset and never pools datasets. The population is every eligible held-out parcel, including non-delivery. The analysis unit is the parcel; dependence is clustered by `patch_id`. The class universe is fixed from training, and the complete operating point is fixed from training and validation. Below 3 paired clusters or 3 paired spatial blocks, respectively, the corresponding view is descriptive: it has no interval, no p-value, and no Holm decision. No transport claim is made.

Complete loss space. Correct precise delivery is the zero-loss reference `R0`; the 6 cost-bearing cells are `L1`; `L2`; `L3`; `L4`; `L5`; `L6`. Removing only common positive scale produces a 5-dimensional simplex. The empirical object is each arm's outcome-frequency vector. Uniform dominance is componentwise; winner regions are exact linear half-spaces and pairwise switching boundaries are hyperplanes conditional on the remaining losses. No elicited point or affected-party preference is required or claimed.

Frozen inference. The primary family uses family-wise alpha 0.05, 9999 paired patch-cluster resamples, and seed 20260909. Within each confirmatory stratum its maximum statistic covers 15 arm pairs by 6 cost coordinates, or 180 directional inequalities. Target power is 0.8 under 999 prospective simulations and a one-sided Wilson admission bound. The nominal intervention point uses training mass 0.9, abstention delivery 0.9, adaptive miscoverage 0.1, and precise backoff 0.9. H2 is secondary and uses the class-unweighted mean absolute deviation of promise retention in one Holm family of two. These are author choices, not data constraints or empirical findings.

Predictor panel. The predictor is a sensitivity factor, not a selected winner. The panel contains 5 members from 4 families; the declared minimum is 3. Selection of a single predictor is allowed only through nested selection independent of evaluation.

Member	Family	Temporal
`unet`	`cnn_encoder_decoder`	no
`deeplabv3plus`	`cnn_atrous`	no
`segformer`	`transformer_espacial`	no
`tsvit-pheno`	`transformer_temporal`	yes
`tsvit-pheno-fullm`	`transformer_temporal`	yes

Set-valued arm. The set-valued action uses deterministic, non-randomized split-conformal adaptive prediction sets. Calibration uses validation labels only. The score is cumulative probability through the true label; prediction returns the smallest nonempty probability-ranked prefix that reaches the calibrated mass, with stable class index as the tie-break. An unattainable finite-sample rank returns the complete catalogue containment over all held-out parcels.

Hierarchical arm. The fixed HCAT4 cut maps 18 semantic leaves to 7 coarse actions. The table publishes the entire cut; parenthetical 'approx.' marks a semantic correspondence that is not exact, and every unmarked correspondence is exact. Confidence for delivery is fixed from training and validation only.
`3310100000` **cereal**. Fine labels: Soft winter wheat; Corn; Winter barley; Spring barley; Winter triticale; Winter durum wheat; Mixed cereal; Sorghum.
`3310200000` **legumes**. Fine labels: Leguminous fodder *(approx.)*; Soybeans.
`3310300000` **potatoes**. Fine labels: Potatoes.
`3310604000` **oilseed crops**. Fine labels: Winter rapeseed; Sunflower.
`3310700000` **fresh vegetables**. Fine labels: Beet; Fruits, vegetables, flowers *(approx.)*.
`3320000000` **grassland grass**. Fine labels: Meadow *(approx.)*.
`3330000000` **permanent crops perennial**. Fine labels: Grapevine; Orchard.

Spatial partition. Selection uses a construction buffer of 1.0 km and requires a minimum centroid-separation ratio of 10.0. The rule is unique eligible candidate or stop; class support is diagnostic, not a selection input. The complete sensitivity grid is 5; 8; 10; 12; 15; 20; 25. These overlapping partitions are sensitivity views, not independent replications.

Class-imbalance sweep. The outcome-free grid contains 20 targets: catalogue sizes 3; 4; 5; 6 crossed with support exponents 0.0; 0.25; 0.5; 0.75; 1.0, at taxonomy depth 1 and minimum training support 100. The common semantic support has 6 classes (wheat, barley, rapeseed, corn, meadow, orchard). The producer has exactly 6 declared outputs, all weights, effective sample sizes, targets, or provenance; the contract forbids inference and any empirical execution before the signed preregistration.

Confirmatory-dataset search boundary. The versioned register contains 14 adjudicated candidate entries through 2026-09-08 and no admitted confirmatory dataset. The dataset search did not preserve a query log, so this is a bounded negative result over the recorded entries, not a globally exhaustive claim. Eligibility is an author-chosen scope and reproducibility rule, not an intrinsic constraint on crop mapping. PASTIS and BreizhCrops remain exploratory. Reopening requires a versioned amendment committed before any new candidate is reviewed.

Model identities. Aliases are not provenance. The complete frozen identity tuple records temporal length, embedding dimension, training and validation folds when known, and the checkpoint digest. Missing folds remain 'not recorded' rather than being inferred from a held-out flag. `tsvit-pheno`: $T = 10$, dimension $= 128$; training folds 1;2;3, validation folds 4;
checkpoint MD5 `b69582e3b55bb0fe8f4aa5d070a3a424`.
`tsvit-pheno-fullm`: $T = 37$, dimension $= 192$; training folds 1;2;3, validation folds 4;
checkpoint MD5 `22ae8074b2c689565f8b854a56e4c5f5`.

`tsvit-pheno-fullm-v2`: $T = 32$, dimension $= 192$; training folds not recorded, validation folds not recorded;
checkpoint MD5 `4a8d83d3fd0b593401725dbfbe2c3e22`.

References

1. Barriere, V., Claverie, M., Schneider, M., Lemoine, G., d'Andrimont, R.: Boosting crop classification by hierarchically fusing satellite, rotational, and contextual data. Remote Sensing of Environment **305**, 114110 (2024). https://doi.org/10.1016/j.rse.2024.114110
2. Carvalho, T., Martinez, J.A.C., Oliveira, H., dos Santos, J.A., Feitosa, R.Q.: Outlier exposure for open set crop recognition from multitemporal image sequences. IEEE Geoscience and Remote Sensing Letters **20**, 1–5 (2023). https://doi.org/10.1109/LGRS.2023.3244532
3. Chow, C.: On optimum recognition error and reject tradeoff. IEEE Transactions on Information Theory (1970). https://doi.org/10.1109/tit.1970.1054406
4. Chzhen, E., Denis, C., Hebiri, M., Lorieul, T.: Set-valued classification — overview via a unified framework (2021)
5. Geifman, Y., El-Yaniv, R.: Selective classification for deep neural networks. In: Advances in Neural Information Processing Systems 30 (NIPS 2017). pp. 4878–4887. Curran Associates (2017), https://proceedings.neurips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html
6. Geifman, Y., El-Yaniv, R.: SelectiveNet: A deep neural network with an integrated reject option. In: Proceedings of the 36th International Conference on Machine Learning (ICML). Proceedings of Machine Learning Research, vol. 97, pp. 2151–2159. PMLR (2019), https://proceedings.mlr.press/v97/geifman19a.html
7. Ghassemi, B., Izquierdo-Verdiguier, E., d'Andrimont, R., Vuolo, F.: How accurately and in what detail can land use and land cover be mapped using copernicus Sentinel and LUCAS 2022 data? Remote Sensing **17**(8), 1379 (2025). https://doi.org/10.3390/rs17081379
8. Gimenez, R., Laloue, A., Fabre, S.: Rejection methods for vegetation mapping using hyperspectral airborne data. International Journal of Remote Sensing **44**(16), 4937–4962 (2023). https://doi.org/10.1080/01431161.2023.2240520
9. Gonzalez-Calabuig, M., Cohrs, K.H., Nedungadi, V., Osika, Z., Cartuyvels, R., Knoblauch, S., Massant, J., Nath, S., Ebel, P., Sitokonstantinou, V.: SHRUG-FM: Reliability-aware foundation models for Earth observation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops. pp. 7919–7928 (2026)
10. Ha, T.M.: The optimum class-selective rejection rule. IEEE Transactions on Pattern Analysis and Machine Intelligence **19**(6), 608–615 (1997). https://doi.org/10.1109/34.601248
11. Jones, E., Sagawa, S., Koh, P.W., Kumar, A., Liang, P.: Selective classification can magnify disparities across groups. In: 9th International Conference on Learning Representations (ICLR) (2021), https://openreview.net/forum?id=N0M_4BkQ05i
12. Liu Ziyin, Wang, Z., Liang, P.P., Salakhutdinov, R., Morency, L.P., Ueda, M.: Deep gamblers: Learning to abstain with portfolio theory. In: Advances in Neural Information Processing Systems 32 (NeurIPS 2019). pp. 10622–10632. Curran Associates (2019), https://proceedings.neurips.cc/paper/2019/hash/0c4b1eeb45c90b52bfb9d07943d855ab-Abstract.html

13. Mortier, T., Hüllermeier, E., Dembczyński, K., Waegeman, W.: Set-valued prediction in hierarchical classification with constrained representation complexity. In: Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence. Proceedings of Machine Learning Research, vol. 180, pp. 1392–1401. PMLR (2022), https://proceedings.mlr.press/v180/mortier22a.html
14. Mortier, T., Wydmuch, M., Dembczyński, K., Hüllermeier, E., Waegeman, W.: Efficient set-valued prediction in multi-class classification. Data Mining and Knowledge Discovery **35**, 1435–1469 (2021). https://doi.org/10.1007/s10618-021-00751-x
15. Pugnana, A., Ruggieri, S.: AUC-based selective classification. In: Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS). Proceedings of Machine Learning Research, vol. 206, pp. 2494–2514. PMLR (2023), https://proceedings.mlr.press/v206/pugnana23a.html
16. Rey, M., Mnih, A., Neumann, M., Overlan, M., Purves, D.: Uncertainty evaluation of segmentation models for Earth observation (2025)
17. Turkoglu, M.O., D'Aronco, S., Perich, G., Liebisch, F., Streit, C., Schindler, K., Wegner, J.D.: Crop mapping from image time series: Deep learning with multi-scale label hierarchies. Remote Sensing of Environment **264**, 112603 (2021). https://doi.org/10.1016/j.rse.2021.112603